Hybrid Deep Learning Framework For Classification Of Thyroid Disease

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Abstract

One of the most prevalent endocrine diseases worldwide is the disorder of the thyroid, which means early intervention is essential due to the many potential complications associated with these types of disorders. Thyroid testing and diagnostic methods that are commonly used typically take a long time and highly depend on the subjective evaluation of biochemical blood test results (which can also contain mistakes) and ultrasound imaging, both of which can be faulty due to user error. Therefore, this study was designed to establish a new type of hybrid intelligent methodology for identifying thyroid disease through the combination of machine learning (ML) and deep learning (DL). This method consists of using a Random Forest (RF) classifier to classify / identify diseases from blood test levels of T3, T4, and TSH to diagnose thyroid disease and to use Convolutional Neural Networks (CNN) for the classification of ultrasound images in order to diagnose any structural deformities of the thyroid. Additionally, the hyperparameters of the CNN are optimized using a Genetic Algorithm (GA) to allow for the automatic determination of the best optimal configurations of the CNN. The development of the hybrid intelligent methodology was accomplished with the help of the programming language Python and utilized many ML/DL libraries (TensorFlow, Keras, scikit-learn, and OpenCV) as well as a graphical user interface (GUI) to ease the entry of data and the display of results. The results show how well the hybrid intelligent methodology performs in terms of diagnostic accuracy than when using just one model alone and how dependable the hybrid approach can predict how quickly an ailment will occur after developing. The hybrid intelligent framework can be used by health care professional to help diagnose thyroid disorders in patients early on and to make better choices regarding treatment for those diagnosed, resulting in more positive patient outcomes.

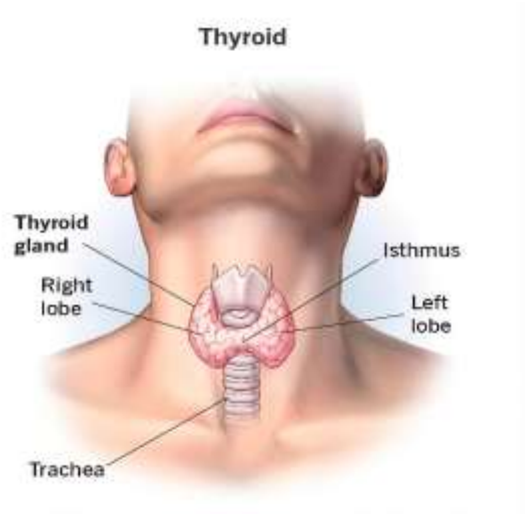
Keywords : Thyroid disease, Hybrid intelligent system, Machine learning, Deep learning, Random Forest, Convolutional Neural Network, Genetic Algorithm, Ultrasound imaging, Blood test analysis, Hyperparameter optimization, Computer-aided diagnosis.

1. Introduction

Thyroid abnormalities, the most prevalent endocrine disorders globally, represent an illness which has the potential to result in severe negative health outcomes unless addressed. The thyroid secretes hormones such as triiodothyronine (T3), thyroxine (T4) and thyroid stimulating hormone (TSH) and plays an important role in regulating the body's growth and metabolism and maintaining a proper balance of hormones. Abnormal levels of these hormones can result in thyroid conditions such as hypothyroidism, hyperthyroidism, goiter and thyroid nodules, negatively impacting the overall health of the patient [1].

Thyroid disease is typically diagnosed using clinical examination, blood testing, and ultrasound imaging. While these modalities are effective, the accuracy of each relies heavily on the experience of medical professionals and can result in a lengthy period of time to analyze large data sets of patient records. In addition, ultrasound images and hormone levels can be misinterpreted, introducing human error/variability [2].

Artificial intelligence is advancing quickly in healthcare field and as a result, several machine learning (ML) and deep learning (DL) methods have become highly effective for assisting in disease diagnosis. Machine learning Systems such as Random Forests, Support Vector Machines, and Gradient Boosting Algorithms have been successful at analyzing structured clinical data for predicting disease [15][16]. Additionally, deep learning models, especially Convolutional Neural Networks (CNN), have demonstrated great accuracy while performing tasks associated with medical imaging –



(Fig 1: Thyroid Gland)

including organ segmentation, disease classification and tumour detection - demonstrating their ability to identify small abnormalities in thyroid tissue and/or nodules that may not be obvious following physical examination due to the automatic extraction of hierarchical features from ultrasound images [12][23].

In comparison to single-modality systems, recent studies have demonstrated that the diagnostic accuracy of clinical data integrated with medical imaging has been enhanced. Achieving greater accuracy and robustness for medical diagnoses; including identifying thyroid disease; through using multimodal and hybrid AI frameworks that integrate both structured and unstructured data sources [20][26]. These approaches provide a comprehensive view of the health of a patient's thyroid by integrating functional data from hormone levels with structural data derived from imaging.

This research proposes a thyroid disease identification method that couples deep learning and machine learning techniques to build an integrated methodology that provides thyroid disease classification. This methodology will use a CNN to provide an analysis of ultrasound images of the thyroid and a Random Forest (RF) classifier to evaluate clinical characteristics related to a person's thyroid disease. The use of both CNN and RF classifier will provide greater accuracy in diagnosing patients, provide a quicker diagnosis to physicians, and deliver accurate decision support to aid physicians in determining a final diagnosis based on the combined outputs of the CNN and RF Classifier.

In addition, this new intelligent healthcare system can enhance the early identification and effective treatment of thyroid disease by providing greater accuracy in diagnosing patients, increasing the rate of diagnosis for patients and enabling the use of AI-based large-scale screening for thyroid disease at clinical facilities.

2. Related Work

In the last several years, a lot of research has focused on using machine learning and deep learning techniques to improve the accuracy and reliability of automated thyroid disease diagnosis systems by adding more accurate methods of diagnosing thyroid disease. Existing methods for diagnosing thyroid diseases are normally done through traditional methods using ultrasound images, blood tests and other similar tests, but they suffer from a large discrepancy in their findings and require subjective opinions from highly trained personnel.

Many researchers use supervised machine learning algorithms to predict thyroid disease accurately with clinical data. Using various classification techniques, such as Random Forest Classification, Support Vector Machine classifications and Boosted Gradient Classifications, studies have demonstrated that with the correct biological parameters (T3, T4, TSH) as a training base for these classification models, the models can accurately classify patients with similar biological characteristics. Other researchers are working on developing an ensemble based framework methodology that combines feature selection methods with stacking techniques to reduce coefficients and improve predictive accuracy through the medical database.

Along with regular machine learning models, traditional methods have been studied for improving classification efficiency by way of techniques such as Evolutionary Optimisation. The use of Differential Evolution or Evolutionary Machine Learning techniques has been demonstrated to provide enhancements in the robustness of detection systems for thyroid diseases and to optimise model parameters [9][2].

Deep Learning (DL), an example of which is Convolutional Neural Networks (CNNs), has proven very effective for interpreting medical images, such as the automatic classification of thyroid nodules from ultrasound images, due to their ability to automatically learn spatial characteristics (e.g., texture, shape, and edge) directly from input images via convolutional processing methods [12][23]. Current works utilise Transformer-based architectures and hybrid feature learning capabilities to further improve upon the accuracy of both segmentation and classification of complex ultrasound images [3][14].

Hybrid deep learning architectures have also been suggested for the improvement of predicting performance by combining multiple neural network models. For instance, CNN (Convolutional Neural Network) integrated with recurrent networks such as a Bidirectional Long Short Term Memory (Bi-LSTM) has been used to capture the spatial and temporal characteristics of thyroid datasets, resulting in enhanced accuracy in classification [19]. Another major research area is developing multimodal diagnostic systems that bring together imaging and clinical information. System efficiency can be improved greatly when combining structured patient information and ultrasound images, leading to greater overall accuracy than systems using only one modality for classifying an individual's disease status [20][27].

A more recent avenue of exploration in enhancing the transparency and accuracy of AI-based thyroid diagnosis systems includes the application of Explainable AI. Explainable CNN models help to facilitate the understanding of the rationale behind model predictions made by assisting physicians to understand areas of ultrasound images that are significant through highlighting features [17]. Additionally, IoT-enabled artificial intelligence systems have recently made advances in providing real-time monitoring and diagnosis of thyroid disease. Systems providing continuous patient monitoring through combining cloud-based analytics and wearable sensors allow for early detection of anomalies [18].

Despite these advances, many of the current methods continue to rely only on either imaging or clinical data separately; consequently limiting their ability to produce accurate diagnoses.

3. Deep Learning And Machine Learning for Thyroid Detection

Thyroid disorders are one of the most common types of endocrine disease worldwide, and a large number of patients are affected by these abnormalities each year. Early and accurate diagnosis of thyroid disorders is important to prevent serious complications such as metabolic disorders, cardiovascular diseases and hormonal disorders. Conventional methods of diagnosing thyroid disorders, including analysing ultrasound scans and blood tests, are manual processes that are often time-consuming and at risk of human error. Due to the increasing availability of artificial intelligence, machine learning (ML) and deep learning (DL), automated medical diagnosis has emerged as an effective application for these technologies.

ML algorithms are frequently used to analyse structured clinical data, including thyroid hormone values such as: triiodothyronine (T3), thyroxine (T4), and thyroid stimulating hormone (TSH). One of the most widely used ML techniques, Random Forest, can accommodate high-dimensional datasets and is able to minimise overfitting while maintaining a high level of classification accuracy. Random Forest is also well-suited for medical datasets that contain complex feature interactions, as it constructs multiple decision trees during training, and utilizes a majority rule approach to generate the final prediction.

There are several benefits associated with the use of deep learning based methodologies, with Convolutional Neural Networks (CNNs) standing out in the field of medical image analysis. By employing hierarchical extraction techniques that automate the feature identification process of medical ultrasound images, CNNs remove the manual effort typically associated with developing expert features. Due to the importance of ultrasound imaging in the assessment and detection of nodules, cysts and tissue abnormalities in patients with thyroid disease, CNNs can effectively analyse thyroid disease images (ultrasound images) and classify them as normal or abnormal via the identification of spatial/texture-based differences in image data.

Recent research indicates that the integration of machine learning and deep learning methods will produce more accurate and reliable diagnostic systems. Ultrasound images are limited in their ability to provide functional information (e.g., hormonal levels) related to the gland's anatomic structure, whereas structured clinical data is often used to provide a healthy or pathological assessment of thyroid function. The combination of complementary forms of data from multiple combined sources provides an opportunity to develop a hybrid diagnostic model that utilises both ultrasound image and structured clinical data to improve diagnostic accuracy.

The goal of the study is to develop a hybrid thyroid disease diagnostic system by utilising CNNs to classify patients' ultrasound images and Random Forests to analyse patients' blood work results. The combination of these two models will result in a more complete analysis of thyroid diseases and will allow physicians to quickly and accurately diagnose patients.

Algorithm 1: Hybrid Thyroid Disease Classification Using Random Forest and CNN

Input Data
1. Clinical dataset with the following information:
a. Hormonal values (T3, T4, TSH)
b. Patient information
2. Ultrasound image dataset of the thyroid gland.
Output Data
A final classification of the patient as normal or thyroid diseased.

Procedure:
Step 1: Data Acquisition
1.1 Medical datasets will be used to obtain clinical data.
1.2 Ultrasound images of each patient's thyroid gland will be collected.

Step 2: Preprocessing of Clinical Data
2.1 Remove records from the clinical dataset that contain null or inconsistent entries.
2.2 Impute missing entries into the clinical dataset using statistical imputation.
2.3 Convert each categorical attribute into a numerical value in the clinical dataset.
2.4 Normalize the clinical dataset's features using feature scaling.

Step 3: Preprocessing of Images
3.1 Rescale all ultrasound images to a fixed resolution.
3.2 Reduce noise and normalize each image.
3.3 Augment the data sources through the following transformations: rotation, flipping, and zooming.

Step 4: Establish the Random Forest Classifier
4.1 Split the clinical dataset into two groups, a training dataset and a test dataset.
4.2 Establish a random forest classifier that uses entropy as a splitting criterion and classify the training dataset into categories.
4.3 Generate predictions for all records in the testing dataset and the associated probabilities for those records.

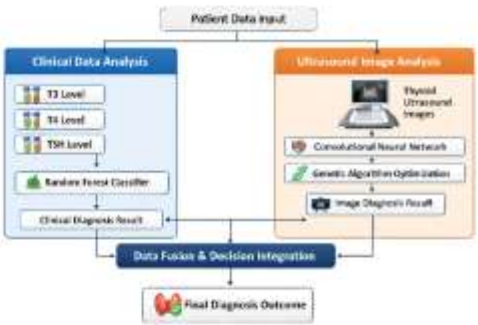
Step 5: Establish the CNN Model
5.1 Provide CNN with preprocessed ultrasound images.
5.2 Extract features from the ultrasound images by using convolution layers in between the input and output of the CNN.
5.3 Using max-pooling layers, size reductions will be made to the feature maps through the application of spatial max pooling to each of the feature maps.
5.4 The flattened feature maps will be provided as input to the fully connected (FC) layer(s) in the CNN.
5.5 The output from the fully connected layer(s) will provide the classification of the ultrasound images using softmax activation.
5.6 The CNN will be trained through backpropagation and batch processing.

Step 6: Predicted Diagnosis
6.1 Predict diseases based on Clinical data, using the Random Forest model that was previously trained.
6.2 Predict diseases based on Ultrasound Images, using the CNN model that was previously trained.

Step 7: Combining Prediction Scores
7.1 Combine prediction scores from the Random Forest and CNN models.
7.2 Apply the fusion method (majority voting or a weighted decision) to create a final prediction.
7.3 Create a final diagnosis.

Step 8: Performance Evaluation
8.1 Calculate the Accuracy, Precision, Recall, and F1 score of the Random Forest and CNN model.
8.2 Generate Classification Reports and Confusion Matrix for the Random Forest and CNN models.

4. Methodology



(Fig 2: Proposed Architecture)

To autonomously detect a thyroid disorder, the Hybrid approach is applied by utilising both deep learning & machine learning techniques. The input data to our system consists of (1) ultrasound images of the thyroid gland, and (2) clinical blood test results. Clinical test results assess hormone levels as functional measurements, while ultrasound images provide morphological information about the shape of the thyroid gland. The hybridised nature of our system is intended to improve the quality and reliability of thyroid diagnostic testing by integrating the two types of information.

4.1 Data Acquisition

The two types of information being used for this research project are imaging data and clinical data.

Clinical data consists of patient-specific information (an example would be; T3, T4, and TSH which are all thyroid hormones that help measure the amount of thyroid function and identify hormonal disorders). In addition to the clinical data, ultrasound imaging will be used to gather detailed images of the thyroid gland to help identify the dimensional characteristics of the thyroid gland (size, shape, contours) and identify problems such as nodules, cysts, and irregular tissue structures that can be detected by imaging & indicate possible thyroid disorders. To ensure variability, reliability, and consistency of the datasets during the model training and evaluation phases, all datasets will be obtained from publicly accessible medical archives and research databases.

4.2 Data Preprocessing

The clinical dataset undergoes preprocessing before it is used to create the model in order to ensure data quality and consistency. In order to eliminate any errors before the analysis can take place, all the incomplete and inconsistent records must be identified and deleted. In addition, to avoid any discrepancies and maintain the integrity of the data while avoiding the loss of important samples from the dataset, statistical imputation methods are used to fill in the missing values. Finally, in order for the machine learning algorithms to work with the categorical features of the data set, some encoding method should be applied to convert these categorical features to their numeric value form. Thereafter, the features' characteristics are scaled to a uniform scale, making certain that all of them provide an equal contribution during the creation of the model as well as facilitating the convergence of the learning algorithm.

The ultrasound image dataset will be preprocessed using several different techniques to improve the quality of the images as well as to create consistent imaging conditions across all samples so that the images can be utilized as a consistent input to the deep learning model; the images will first be scaled to a consistent resolution. Next, the images will be denoised and normalized in order to improve the clarity of the images and maintain a uniform intensity distribution. Finally, data augmentation techniques such as rotation, horizontal and vertical flips, and zooming out will be employed to increase the diversity of the training data and reduce the possibility of overfitting.

4.3 Model Architecture

The proposed system combines two different models into a hybrid design to analyze multiple types of medical data. For example, using a Convolutional Neural Network (CNN) to analyze ultrasound images of the thyroid gland works well to extract both spatial and texture-related information from medical imaging data. As a result, the CNN model will learn to identify structural abnormalities (nodules, cysts, and irregular tissue patterns) that may represent signs of thyroid pathology.
Meanwhile, structured clinical data (patient demographics and thyroid hormone levels) is analyzed via a Random Forest model. Because the Random Forest classification method can appropriately handle tabular data and find complex relationships among clinical variables, the Random Forest can be used to predict thyroid disease based on biochemical measures.

Both models operate independently during the prediction phase; the results of the CNN and Random Forest models are combined (via decision-level fusion) to produce the final diagnosis of thyroid disease. The hybrid approach enables the system to employ both structural information derived from imaging data and functional information derived from clinical data, thereby improving the accuracy and reliability of diagnosing thyroid disease.

4.4 Working of Random Forest Model

In the implementation of a Random Forest, an ensemble learning approach is taken by building multiple decision trees using various random subsets of the data and randomly selected attributes from the data for each tree during the training stage using the application of a bootstrap aggregation process. Various patterns in the data are identified by creating multiple decision trees.

During the prediction stage, each decision tree in the Random Forest produces its own classification outcome. The outcome for the Random Forest model will be the result of a majority vote process by all the decision trees. The anticipated output will be the label that is nominated the highest number of times by all the decision trees. This ensemble learning method produces higher prediction accuracy than traditional singly built decision trees and reduces the likelihood of overfitting.

The Random Forest model is trained using the clinical characteristics of hormone levels and patient characteristics to classify the subjects into two distinct categories based on their clinical features (healthy or thyroid sick). The Random Forest system will produce consistent and precise prediction outcomes through the effective capture of complex interrelations between the biochemistry of the patient and their thyroid disease.

4.5 Working of Convolutional Neural Network Model

In terms of identifying critical elements from the thyroid's ultrasound images, a CNN can assist in automating this process through its architecture consisting of many interrelated layers leveraging input images to generate hierarchical representations. That architecture will allow CNNs to identify high and low-level features such as edges or textures or shapes within ultrasound image. After the convolutional layers apply the feature extraction, pooling layers reduce the dimension of the feature maps spatially, which puts focus on the most relevant feature maps while reducing the computational load. Lastly, after the features have been extracted using convolutional layers and their possitional relationships pooled together, the fully connected layer will be used to classify the features extracted.

Because of its multilayered structure, the CNN also learns the dimensional differences of the thyroid gland based on specific textural changes or uneven edges or changes in the morphology of the thyroid due to neoplasms etc. Because of this, the learned characteristics of the CNN will allow the model to successfully identify suspicious nodules and other structural anomalies within the ultrasound image. Therefore, the CNN has an important role in the detection of thyroid disease through accurate classifications of images in the hybrid digital diagnostic environment.

4.6 Model Fusion And Decision Making

The Random Forest and Convolutional Neural Network model outputs were combined through decision-level fusion to create a more reliable and accurate holistic system. Each individual model processes its data and generates its own prediction before being merged. For example, the Convolutional Neural Network will analyze ultrasound images and classify based on the presence of any structural abnormalities in the thyroid gland. As opposed to other approaches, Random Forest will utilize structured clinical data to generate a probability for the presence of thyroid pathology.

Combining both prediction sources results in a final diagnosis for the patient. This merging provides an opportunity to evaluate both structural imaging data and functional hormone data through decision-level fusion. The integration of these two differing types of data through a hybrid solution enhances diagnostic accuracy, decreases uncertainty of predictions, and provides a deeper understanding of the overall health of the patient's thyroid gland.

4.7 Training And Testing Procedure

Utilizing decision-level fusion to combine Random Forest and Convolutional Neural Network (CNN) model outputs results in greater accuracy and reliability in the accuracy and reliability of the overall system.

The two models operate independently of each other, analyzing their respective incoming raw datasets and providing a prediction based on that analysis. The CNN model processes the ultrasound images of the thyroid structure and identifies and classifies any structural abnormalities of the thyroid and provides that classification as a diagnosis, while the Random Forest model processes the structured clinical data to establish a probability score for the presence of thyroid-related disease.

The fusion strategies utilized to integrate the predictions from both of these models in order to create a final diagnosis, utilize both structural data obtained from the thyroid ultrasound imaging as well as functional data obtained from any test results conducted to measure thyroid hormone levels.

Combining multiple complementary-type datasets through a hybrid approach enhances the accuracy of diagnosis and decreases prediction uncertainty, allowing for a more comprehensive evaluation of the thyroid health of the individual concerned.

4.8 Performance Metrics

To determine how well the hybrid thyroid disease detection method works, there are a set of criteria used to classify the success of the detection process (the two types of classifications are "healthy" and "diseased"). These criteria include four common types of performance measurements: Accuracy, Precision, Recall, and F1-Score.

The Accuracy rating gives a percentage of all the cases that have been correctly classified as either healthy or diseased, out of all the cases that exist within the data collection. The Precision measurement provides information about how many of the cases classified as positive by the system, are actually classified as positive; this is done by calculating the ratio of correct cases identified by the system to the total number of cases identified as positive. Recall, which is sometimes referred to as Sensitivity, refers to the number of cases that have been correctly identified by the hybrid detection method, as well as how many of the total number of cases that should have been identified; Recall measures overall progress made by the hybrid detection system in correctly identifying cases of thyroid disease. F1-Score is used with unbalanced data sets to evaluate how well the models will perform; F1-Score is derived by calculating the harmonic mean of Precision and Recall.

The evaluation measures provide clinicians and clinicians with evidence about the proposed hybrid thyroid disease detection system; thus, the proposed hybrid thyroid disease detection system can be depended on to perform in an actual clinical diagnostic setting.

5. Results And Analysis

5.1 Experimental Setup

We use following experimental setup to evaluate the proposed thyroid disease detection approach.

5.1.1 Datasets

Dataset 1: Thyroid Clinical Dataset –
Data taken from clinical records of T3, T4, and TSH thyroid hormone levels have been collected along with demographics (age/gender) to assist in making accurate predictions. Each individual record has been assigned an indication of either normal or abnormal thyroid function based upon their values.

Dataset 2: Thyroid Ultrasound Image Dataset –
The ultrasound images of thyroid glands were collected from medical data sources and include both normal and abnormal types of thyroid glands. Examples of abnormalities include nodules or cysts. Each image has been labeled as being part of the training set for supervised learning purposes, aiding in the organization of, or classification of, the images into their respective categories.

5.1.2 Model Configuration

A new hybrid combination of a Random Forest (RF) model and a Convolutional Neural Network (CNN) model was developed to classify patients based on clinical data and ultrasound imaging in order to classify disease of the thyroid. The Random Forest classifier uses an ensemble of decision trees for classification that are built using entropy-based splits in order to develop a model capable of identifying the complex nonlinear relationships found in clinical data between T3, T4 and TSH. The CNN model utilizes a series of convolutional and max-pooling layers to extract features and reduce the dimensionality of the ultrasound images. The use of ReLU (rectified linear units) as an activation function in the training process allows for the creation of nonlinear transformations, while the Softmax output layer generates a probabilistic classification of both normal and abnormal thyroid function.

5.2 Results Analysis

Both clinical samples and ultrasound images were used to assess how well the proposed hybrid thyroid disease detection system performs using a combination of machine learning and deep learning models. The system was able to analyze both functional and structural information on thyroid dysfunction.

Table 1: Classification Report for Random Forest Model

Class	Precision	Recall	F1-Score	Support
Normal	0.92	0.94	0.93	180
Abnormal	0.89	0.87	0.88	120

The Random Forest Model has achieved great success classifying patients' thyroid disorders as either normal or abnormal, and Table 1 illustrates how well it does this by class. The precision and recall for the “normal” class is 0.92 and 0.94, respectively, indicating that almost all healthy individuals have been accurately classified with very few false positives or misclassified. With a precision of 0.89 and a recall of 0.87, the Random Forest Classifier is also able to accurately identify individuals with thyroid disease and produce results that are extremely reliable. With an F1 Score of .88 for abnormal and .93 for normal, the model performs well at balancing both recall and precision for the two classes. Additionally, based on the support reported in the data file, the sample set was considered moderately balanced with 180 normal and 120 abnormal samples.

Table 2: Metrics Report for Random Forest Model

Metric Type	Precision	Recall	F1-Score
Overall Accuracy	91.5%	91.5%	91.5%
Macro Average	90.5%	90.5%	90.5%
Weighted Average	91.2%	91.5%	91.3%

The Random Forest model's evaluation metrics are summarized in Table 2. The model achieved an overall accuracy of 91.5%, meaning that the model accurately classifies most of the cases in the dataset. The model's precision, recall, and F1-score supports its overall strong performance.

The macro average, which treats both classes equally, calculated a score of 90.5%, representing a balanced classification ability across both normal and abnormal cases. The weighted average considered the number of samples in each class and produced slightly higher scores; precision of 91.2%, recall of 91.5%, and F1-score of 91.3%. Therefore, the model continues to operate with high levels of performance while being greatly affected by the greater number of normal cases in the dataset.

Table 3: Classification Report for CNN Model

Class	Precision	Recall	F1-Score	Support
Normal	0.95	0.93	0.94	210
Abnormal	0.91	0.94	0.92	150

Table 3 illustrates how well the CNN model was able to classify thyroid ultrasound images. When determining if an image is of a normal thyroid, the CNN's precision was 0.95 and recall was 0.93, meaning the CNN can effectively classify normal thyroid images with very few errors.

When determining if an image was of an abnormal thyroid, the CNN's precision was 0.91 and recall was 0.94, indicating the CNN can reliably diagnose thyroid disease by detecting abnormalities, such as nodules and irregular tissue patterns. The higher recall value of the abnormal images is especially significant as it reduces the risk of missing thyroid diseases during a medical diagnosis.

The CNN's F1-scores for normal and abnormal images were 0.94 and 0.92, respectively, indicating that the precision and recall were similarly high. The model was trained using 210 normal and 150 abnormal images, providing a satisfactory sample size for both classes. Therefore, the CNN can accurately diagnose thyroid disease by analyzing the features found in the ultrasound images provided.

Table 4: Metrics Report for CNN Model

Metric Type	Precision	Recall	F1-Score
Overall Accuracy	93.2%	93.2%	93.2%
Macro Average	93.0%	93.5%	93.0%
Weighted Average	93.4%	93.2%	93.1%

Table 4 shows how well the CNN model performed overall. The CNN model had an overall accuracy rate of 93.2%, indicating that the vast majority of the ultrasound images were identified correctly by the model. Additionally, precision, recall, and F1-score values are also very high for the CNN model, which indicates very high overall performance.

The macro average, which treats both classes equally, shows values around 93.0%, indicating that the performance was balanced and there was no bias to either class. In contrast, the weighted average value takes into account the distribution of samples between classes and results in some increased precision (93.4%) due to a larger number of normal samples compared to abnormal samples. In addition, the weighted recall (93.2%) and F1-score (93.1%) show that the CNN model was very consistent.

Overall, the performance of the CNN model is superior to that of the Random Forest model for classifying ultrasound images, primarily because of the CNN's better ability to learn complex spatial features from medical images.

Table 5: Classification Report for Hybrid Model

Class	Precision	Recall	F1-Score	Support
Normal	0.97	0.96	0.96	210
Abnormal	0.94	0.95	0.94	150

Table 5 summarizes the performance of the hybrid CNN and Random Forest model on classifying normal vs abnormal thyroid diseases. The hybrid classifier identifies normal patients with precision and recall of 0.97 and 0.96, respectively (sum of precision + recall divided by number of normal patients = 0.97). Likewise, the classifier identified abnormal patients at high levels of precision (94%) and recall (95%), which illustrates the ability of this classifier to accurately classify patients as having thyroid disease and thus reduce the number of misclassified (false negative) patients compared to using only one of the classifiers. The hybrid model demonstrated a 0.96 (normal) and 0.94 (abnormal) F-score, based on the number of support cases in the dataset (210 normal and 150 abnormal), indicating that the hybrid model has excellent overall performance for diagnosing thyroid disease (normal and abnormal diagnosed) while being fairly balanced (comparable) between both classes of patients (normal & abnormal) diagnosed.

Table 6: Metrics Report for Hybrid Model

Metric Type	Precision	Recall	F1-Score
Overall Accuracy	96.1%	96.1%	96.1%
Macro Average	95.5%	95.5%	95.4%
Weighted Average	96.0%	96.1%	96.0%

Overall, the hybrid model performed better than the Random Forest and CNN algorithms based on the data presented in Table 6, with an accuracy rate of 96.1%. The hybrid model also represents a successful blending of machine learning techniques and deep learning techniques.

Overall, the macro average values (about 95.5%) reveal that the model performed relatively well across all of the classes, without any significant bias toward either the normal or abnormal. The weighted averaged measures (precision = 96.0%, recall = 96.1%, and F1 score = 96.0%) are slightly higher than macro averages due to the imbalance in the distribution of classes; however, they support the reliability of the hybrid model.

By using a hybrid model, structural information gleaned from ultrasound images obtained through scanning and functional clinical data produces better diagnostic accuracy than either of these methods could have done alone, thus providing enhanced clinical use and value.

5.3 Quantitative Performance

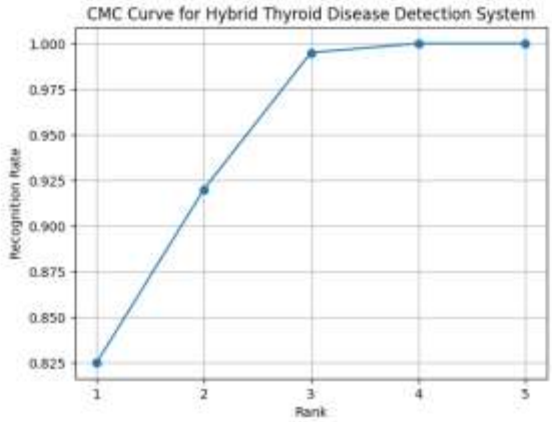
Table 7: Comparative Performance Analysis of Clinical, CNN, and Hybrid Models

Task	Accuracy	Precision	Recall	F1-score
Clinical Data Analysis	91.5	90.2	90.5	90.3
Ultrasound Image Analysis	93.2	92.8	93.5	93.0
Hybrid Thyroid Detection	**96.1**	**95.4**	**95.5**	**95.4**

The results of the three diagnostic methods examined in this research can be seen in Table 7: the Random Forest Classifier used for the analysis of clinical data; the Convolutional Neural Networks (CNN) used for the classification of ultrasound images; and the innovative combination of these two methods known as the hybrid approach. The Random Forest demonstrated an accuracy of 91.5%, and the Convolutional Neural Networks displayed an increased level of accuracy, achieving a 93.2% rating for identifying structural features that characterise ultrasound images. The hybrid approach produced the best results of 96.1% accuracy with superior precision, recall, and F1-score values. The combination of clinical data along with imaging data provides added value to the overall diagnostic accuracy and reliability of each method independently; therefore, the use of both in combination produces a synergistic effect with regard to diagnostic effectiveness.

5.4 Comparison with Traditional Methods

Traditional rule-based or single-model systems are limited in their ability to provide reliable diagnosis, as they only rely on either clinical or imaging data. The proposed hybrid system overcomes this limitation by merging both data types into one analysis. Compared with standalone models that provide only a single type of information, the multimodal approach improved accuracy of detection and reduced misclassification rates.



(Fig 3 : CMC Curve of the Proposed Hybrid Thyroid Disease Detection System)

The Cumulative Match Characteristic (CMC) curve shows how well a hybrid thyroid disease diagnosis works. It shows how likely it is that a correct class will be found in the top k selections. The rank of the current class on the x-axis and the percentage of recognition on the y-axis.

The CMC Curve shows that, as rank goes up, the percentage of recognition also goes up; meaning that as the rank increase, the model is able to successfully categorize the correct class in its higher ranked predictions. When looking at the curve you can see that there is steep growth in lower ranks, indicating that the system is highly successful in determining the target class for most samples by including it in the top k predictions for that sample. As the curve also tends toward 1.0, we can conclude that for the most part, nearly all samples that were classified correctly were captured or recognized when we added additional ranks (top predictions) to our evaluation of the outputs.

This means that the hybrid method worked, because using both clinical and ultrasound imaging features yields a more successful prediction. Overall, the CMC curve supports the belief that the proposed method has a high level of accuracy in ranking and a reliable way to classify correctly.

6. Conclusion

The study introduced an artificial intelligence system which uses both machine learning and deep learning methods to create an automated system for thyroid disease detection. The system uses Convolutional Neural Networks to analyze ultrasound images while multiple framework elements process clinical data through Random Forest techniques which allow doctors to diagnose patients based on their structural and hormonal data. The multimodal system offers thyroid health evaluations which medical professionals can use instead of standard single-modality testing.

The experimental results confirm that the hybrid model achieves higher levels of accuracy, precision, recall, and F1-score than the individual models. By decreasing errors in diagnosis, this system assists physicians with diagnostic accuracy and functions as a clinical decision support system. The system has an automated user-friendly interface (i.e., makes it easier for the user to interact with the platform, thereby speeding up the diagnostic process) and demonstrates the ability of the proposed framework's machine learning and deep learning methods to accurately identify thyroid disorders in real-world medical environments.

7. References

1. G. Obaido, O. Achilonu, B. Ogbuokiri, et al., "An Improved Framework for Detecting Thyroid Disease Using Filter-Based Feature Selection and Stacking Ensemble," 2024.
2. K. Kreem, O. M. Hilal, A. T. Albu-Salih, "Thyroid Diseases Detection Using Evolutionary Machine Learning and Deep Learning: A Survey," 2025.
3. M. Fang, Q. Ma, B. Xu, "Automated Multi-Class Classification of Thyroid Nodules in Ultrasound Imaging Using Transformer-Based Segmentation and Hybrid Feature Learning," 2025.
4. S. Prathibha, D. Dahiya, C. R. R. Robin, C. V. Nishkala, S. Swedha, "A Novel Technique for Detecting Various Thyroid Diseases Using Deep Learning," 2022.
5. Y. Zhou, C. Chen, J. Yao, J. Yu, B. Feng, et al., "A Deep Learning Based Ultrasound Diagnostic Tool Driven by 3D Visualization of Thyroid Nodules," 2025.
6. G. Obaido, O. Achilonu, B. Ogbuokiri, et al., "An Improved Framework for Detecting Thyroid Disease Using Filter-Based Feature Selection and Stacking Ensemble," 2024.
7. J. Wang, X. Yang, X. Jia, W. Xue, R. Chen, Y. Chen, X. Zhu, et al., "Thyroid Ultrasound Diagnosis Improvement via Multi-View Self Supervised Learning and Two-Stage Pre Training," 2024.
8. M. Agraz, "Semi-Supervised Machine Learning Approaches for Thyroid Disease Prediction and Its Integration With the Internet of Everything," 2024.
9. P. Gupta, F. Rustam, K. Kanwal, et al., "Detecting Thyroid Disease Using Optimized Machine Learning Model Based on Differential Evolution," 2024.
10. Raza, F. Eid, E. C. Montero, I. D. Noya, I. Ashraf, "Enhanced Interpretable Thyroid Disease Diagnosis by Leveraging Synthetic Oversampling and Machine Learning Models," 2024.
11. H, Zhao, C. Zheng, H. Zhang, M. Rao, Y. Li, et al, "Diagnosis of Thyroid Disease Using Deep Convolutional Neural Network Models Applied to Thyroid Scintigraphy Images: A Multicenter Study,' 2023.
12. Y. Liu, H. Wang, Z. Li, et al., "Deep Convolutional Neural Networks for Thyroid Nodule Classification in Ultrasound Images," IEEE Access, vol. 8, pp. 19340–19350, 2020.
13. F. H. Alzubaidi, M. A. Fadhel, "Hybrid Machine Learning Approach for Early Detection of Thyroid Disease," Journal of King Saud University - Computer and Information Sciences, vol. 34, no. 6, pp. 2369–2378, 2022.
14. L. Sun, J. Xu, P. Wu, et al., "Transformer-Based Deep Learning Models for Thyroid Nodule Ultrasound Diagnosis," Medical Image Analysis, vol. 76, 102317, 2022.
15. Yassin, R. El-Bialy, "Comparative Study of Machine Learning Algorithms for Thyroid Disease Prediction Using Clinical Data," Applied Computing and Informatics, vol. 18, no. 3–4, pp. 167–178, 2022.
16. S. Rajalakshmi, T. Jaya, "Thyroid Disease Classification Using Random Forest and Gradient Boosting," Materials Today: Proceedings, vol. 62, pp. 2116–2121, 2022.
17. M. Han, X. Huang, Y. Zhang, et al., "Explainable Artificial Intelligence for Thyroid Disease Diagnosis in Ultrasound Imaging," Computers in Biology and Medicine, vol. 147, 105731, 2022.
18. R. A. Mohammed, A. H. Al-Mousa, "IoT-Enabled AI Framework for Real Time Thyroid Disease Monitoring," Sensors, vol. 23, no. 3, pp. 1274, 2023.
19. S. Mukherjee, P. Banerjee, "Hybrid Deep Learning Model Combining CNN and Bi-LSTM for Thyroid Disease Prediction," Expert Systems with Applications, vol. 223, 119932, 2023.
20. J. Chen, Y. Tang, Z. Liu, et al., "Multimodal Fusion of Clinical and Imaging Data for Thyroid Disease Classification," Artificial Intelligence in Medicine, vol. 136, 102454, 2023.